RESEARCH BLUEPRINT

Quantum Error-Correcting Codes from Galois Hulls of Cyclic Codes over Noncommutative Matrix Rings

A phase-by-phase research plan and collaboration protocol for Arena AI and ChatGPT

CORE WORKFLOW

Arena AI develops the research artefacts phase by phase. ChatGPT independently audits the mathematics, literature, proofs, computations and writing, then prepares the next precise Arena prompt. The researcher (Arun) approves every phase gate. No phase is treated as complete merely because an AI system says it is complete.

How to use this file

Is file ko Arena AI aur ChatGPT dono ko dena hai. Dono systems ko isi document ko single source of project instructions maana hai. Arena ka kaam develop karna hai; ChatGPT ka kaam verify, challenge, improve aur next phase ke liye exact prompt banana hai. Researcher final mathematical decisions aur phase approvals dega.

1. Pehle poora blueprint dono systems ko upload karein.
2. Arena ko sirf Phase 0 ka prompt dein. Ek hi baar mein complete paper likhne ko na kahein.
3. Arena ke outputs aur files ChatGPT ko dein. ChatGPT se audit aur next Arena prompt banwayein.
4. ChatGPT ka approved prompt Arena ko dein. Arena agla phase execute kare.
5. Har phase ke baad GO / REVISE / STOP decision lein. Approval ke bina next phase par na jaayein.
6. Final manuscript tabhi assemble karein jab all mathematical claims, references and computational results independently checked hon.

1. Project identity and research objective

Working title: Quantum Error-Correcting Codes from Galois Hulls of Cyclic Codes over Noncommutative Matrix Rings.

Research area: Algebraic Coding Theory; finite rings and matrix rings; cyclic codes; Galois duality and hulls; Gray maps; quantum error-correcting codes.

Central objective: Develop a mathematically valid framework for cyclic codes over a carefully selected noncommutative matrix ring, define an appropriate Galois duality and hull theory, characterize the resulting hulls, and use rigorously justified classical-to-quantum constructions to obtain quantum code families.

The title and scope are provisional. Phase 0 must establish whether the exact intersection is genuinely new and whether the proposed algebraic framework is feasible.

2. Proposed mathematical setting — not yet fixed

Candidate coefficient ring:

```
A_t = F_q[u]/<u^t>,     u^t = 0.
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Candidate matrix ring:

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R = M_m(A_t),
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with candidate values m = 2, 3 or 4 and q = p^r. These are alternatives to investigate, not assumptions to impose. Arena must compare them and recommend a setting only after checking the literature, module structure, automorphisms, inner products and quantum-code route.

Notation rule: use t for the nilpotency exponent and s (or ell) for the Galois automorphism parameter. Do not use k for both.

Potential field automorphism:

```
sigma_s(a) = a^(p^s),     a in F_q.
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Its extension to matrices may be entrywise, but Arena must prove that the extension is compatible with matrix multiplication and specify its action on the full coefficient ring. If q = p, the field automorphism may be trivial; this case must not be presented as a nontrivial Galois setting.

Critical issue: the inner product on a noncommutative matrix ring cannot be assumed. Any proposed pairing must specify its domain and codomain, left/right module convention, additivity, non-degeneracy, automorphism behaviour and compatibility with the code operations. Candidate formulas are hypotheses to test, not established definitions.

3. Research questions

7. Which finite noncommutative matrix-ring family supports a useful and mathematically consistent cyclic-code theory for this project?
8. Can a nontrivial Galois automorphism be defined on the selected ring, and what are its fixed elements and algebraic properties?
9. What inner product or pairing gives a valid Galois dual for the chosen left/right module structure?
10. Under what conditions is the Galois dual of a cyclic code again cyclic?
11. Can Galois hulls be structurally characterized using generator data, parity-check data or suitable matrices?
12. Can exact hull-dimension formulas be derived for meaningful code families?
13. Does a suitable Gray map exist, and what precisely does it preserve: linearity, distance, duality, hulls, or only a weaker relation?
14. Which classical-to-quantum construction is valid under the proved hypotheses, and what new quantum parameters can be obtained?

4. Proposed section-wise manuscript structure

Section 1. Introduction

- 1.1 Background of algebraic coding theory
- 1.2 Quantum error-correcting codes and classical-to-quantum connections
- 1.3 Galois duality and hulls: motivation and known developments
- 1.4 Cyclic codes over noncommutative matrix rings
- 1.5 Critical literature review and research gap
- 1.6 Main contributions and organization of the paper

Research control: State only contributions actually proved in the paper. Do not announce hoped-for results as completed contributions.

Section 2. Preliminaries and Algebraic Foundations

- 2.1 Finite fields and the selected coefficient ring
- 2.2 Structure of the selected matrix ring
- 2.3 Left/right modules and code conventions
- 2.4 Ring automorphisms and their extensions
- 2.5 Linear codes and cyclic shifts over the selected ring
- 2.6 Inner products, duals and required algebraic lemmas

Research control: Define the polynomial variable convention, centrality, cyclic-shift action and left/right ideal or module structure precisely.

Section 3. Galois Duality for Cyclic Codes over Matrix Rings

- 3.1 Galois automorphism on the selected matrix ring
- 3.2 Definition and validation of the Galois pairing
- 3.3 Galois dual codes
- 3.4 Duality properties and cyclicity conditions
- 3.5 Generator/parity-check descriptions and main theorems

Research control: Prove well-definedness and non-degeneracy before using the pairing to define dual codes.

Section 4. Structural Characterization of Galois Hulls

- 4.1 Definition of the Galois hull

- 4.2 Algebraic characterization of the hull
- 4.3 Generator and parity-check descriptions
- 4.4 Hull-dimension formulae
- 4.5 Special cases: trivial, nontrivial, self-orthogonal and self-dual cases

Research control: A proposed rank formula is only a target until derived under explicit hypotheses. Do not transfer field formulas to matrix rings without proof.

Section 5. Construction and Classification of Cyclic Codes

- 5.1 Polynomial/module representation
- 5.2 Factorization and generator structure
- 5.3 Construction of cyclic codes over the selected matrix ring
- 5.4 Classification by relevant algebraic parameters
- 5.5 Galois hull dimensions and worked examples

Research control: If a commutative polynomial quotient is not appropriate, replace it with a rigorously defined skew or module-theoretic formulation.

Section 6. Gray Maps and Classical Image Codes

- 6.1 Construction and definition of a Gray map
- 6.2 Additivity/linearity and image alphabet
- 6.3 Weight relationship
- 6.4 Images of cyclic codes
- 6.5 Relationship between ring hulls and image-code duals/hulls

Research control: State exactly what the map preserves. Do not claim hull preservation unless it is proved.

Section 7. Quantum Error-Correcting Code Constructions

- 7.1 Quantum-code preliminaries and notation
- 7.2 CSS construction and its hypotheses
- 7.3 Galois-hull-based construction theorem(s)
- 7.4 Parameter derivation: length, dimension and distance
- 7.5 New quantum code families and explicit examples
- 7.6 Quantum Singleton bound and defect analysis

Research control: A Galois hull alone does not automatically produce a quantum code. Prove field-linearity of the image and the required orthogonality/containment conditions.

Section 8. Computational Results and Verification

- 8.1 Reproducible computational framework
- 8.2 Ring, automorphism and pairing checks
- 8.3 Code, dual and hull computations
- 8.4 Minimum-distance computation and completeness limits
- 8.5 Quantum parameters and comparison tables

Research control: Distinguish exhaustive computation from heuristic search. Include scripts, versions, seeds and exact input parameters.

Section 9. Comparative Discussion and Novelty Analysis

- 9.1 Comparison with matrix-ring cyclic-code literature
- 9.2 Comparison with Galois-hull literature
- 9.3 Comparison with existing quantum constructions

- 9.4 New theoretical results and parameter improvements
- 9.5 Limitations and scope

Research control: Use verified references and comparable parameter conventions. Do not claim novelty solely because a different ring size or notation is used.

Section 10. Conclusion and Future Research

- 10.1 Summary of proved results
- 10.2 Quantum-code outcomes
- 10.3 Limitations
- 10.4 Future directions

Research control: The conclusion must report only results established in Sections 2–8.

5. Phase-by-phase execution plan

Phase 0 — Literature review, novelty and feasibility

Objective: Build a verified literature map covering (i) cyclic codes over matrix rings, (ii) Galois duals/hulls over rings and code families, (iii) quantum codes from matrix rings, and (iv) quantum codes from Galois hulls.

Required deliverables: Annotated bibliography; comparison matrix; exact definitions used in prior work; candidate research gaps; feasibility report; GO/REVISE/STOP recommendation.

Exit gate: No novelty claim without primary-source verification. Identify the closest papers and explain precisely what they already prove.

Phase 1 — Select ring and establish algebraic foundations

Objective: Compare candidate rings and fix one only after checking automorphisms, ideals/modules, left/right conventions and viable inner products.

Required deliverables: Ring specification; notation sheet; lemmas on automorphism; module conventions; validated pairing or proof that the candidate fails.

Exit gate: Do not proceed if the Galois automorphism is trivial or the pairing cannot be made valid for the intended construction.

Phase 2 — Cyclic-code framework

Objective: Define cyclic codes rigorously over the selected structure and derive useful generator or module descriptions.

Required deliverables: Definitions; structural propositions; examples; explicit small cases; computational checks.

Exit gate: Resolve central variable/polynomial quotient issues and left/right closure.

Phase 3 — Galois duality

Objective: Define and study the selected Galois dual; prove its basic properties and determine when duals are cyclic.

Required deliverables: Duality theorems; proofs; generator/parity-check characterization; counterexamples where expected properties fail.

Exit gate: Every theorem must list assumptions and prove both directions when an equivalence is claimed.

Phase 4 — Galois hull theory

Objective: Characterize C intersect C-perp_s and derive dimension/generator results.

Required deliverables: Hull theorems; dimension formulae; special cases; worked examples; independent proof audit.

Exit gate: This is a central novelty gate. Compare every result with existing hull papers.

Phase 5 — Gray maps

Objective: Construct a suitable map to a finite-field alphabet and determine its exact algebraic and metric properties.

Required deliverables: Map definition; proof of additivity/linearity as applicable; weight results; dual/hull relationship theorem or explicit non-preservation result.

Exit gate: Do not assume Gray maps preserve duals or hulls.

Phase 6 — Quantum constructions

Objective: Select a valid quantum construction (CSS or another justified route) and prove all inclusion/orthogonality and parameter claims.

Required deliverables: Construction theorem(s); full parameter derivations; explicit quantum examples; bound analysis.

Exit gate: Do not infer quantum-code existence from hull dimensions alone.

Phase 7 — Computation and reproducibility

Objective: Implement exact arithmetic and independently verify the algebraic, coding and quantum results.

Required deliverables: Magma/Python source; test logs; parameter tables; distance certificates or stated search limits; reproducibility instructions.

Exit gate: No fabricated outputs. Label exhaustive, bounded and heuristic calculations distinctly.

Phase 8 — Manuscript and submission package

Objective: Write the paper using the approved section structure and only validated results.

Required deliverables: LaTeX source; bibliography; figures/tables; compiled PDF if compiler is available; reproducibility package; final audit report.

Exit gate: No fabricated citations, theorem claims, proofs, numerical results or compilation claims.

6. Collaboration protocol: Arena AI × ChatGPT × Researcher

Participant	Primary responsibility	Must not do
Arena AI	Execute the currently authorized phase; inspect supplied papers/repositories; derive definitions, proofs, examples, code and files; maintain a decision and change log.	Must not skip phases, invent references/results, silently change the ring or notation, or declare unverified claims proved.
ChatGPT	Audit Arena outputs independently; check definitions, assumptions, proof steps, literature overlap, code logic and writing; identify errors; prepare the next phase-specific Arena prompt.	Must not rubber-stamp Arena output, assume tool output is correct, or silently fill missing mathematical arguments.
Researcher (Arun)	Choose research scope; provide source papers; resolve mathematical choices; approve phase gates; decide final claims and authorship/submission decisions.	Must not treat AI-generated novelty or proof claims as verified without review.

7. The mandatory hand-off loop

15. Researcher gives Arena the master blueprint and the prompt for the current phase only.
16. Arena returns its report, source files, proof details, citations, code and unresolved questions. It must list assumptions and any failed attempts.

17. Researcher uploads the complete Arena output to ChatGPT and uses the ChatGPT Audit Prompt in Section 8.
18. ChatGPT returns: (a) verdict, (b) mathematical errors/gaps, (c) literature/novelty concerns, (d) corrections required, (e) exact next Arena prompt. If the phase is not sound, ChatGPT must not advance the project.
19. Researcher sends the audit and approved next prompt to Arena. Arena revises the current phase or begins the next one as authorized.
20. Repeat until Phase 8. Researcher makes the final GO decision for manuscript assembly.

8. Copy-paste master instructions

8A. Master instruction for Arena AI

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You are the research-development agent for the project described in the uploaded
Research Blueprint. Read the entire blueprint before acting.

PROJECT:
“Quantum Error-Correcting Codes from Galois Hulls of Cyclic Codes over
Noncommutative Matrix Rings.”

WORKING RULES:
1. Execute ONLY the phase explicitly assigned in the current prompt.
2. Do not write the complete paper prematurely.
3. Treat the ring, automorphism, inner product and construction as provisional
   until proved mathematically valid.
4. Verify primary literature and provide complete bibliographic details,
   DOI/URL where available, and the exact result relevant to this project.
5. Never invent citations, theorem statements, proofs, computations or files.
6. For every theorem, state all assumptions, give a complete proof, and identify
   dependencies on earlier lemmas.
7. Explicitly distinguish established facts, your proposed definitions,
   conjectures, computational observations and unresolved questions.
8. Maintain consistent notation: t = nilpotency exponent; s or ell =
   Galois parameter. Do not overload k.
9. Explain left/right module conventions, polynomial variable centrality,
   and the exact code/dual definitions.
10. A Gray map or Galois hull does not automatically yield a quantum code.
    Prove the required field-linearity, orthogonality/containment and parameter
    formula before claiming a quantum construction.
11. For computation, provide runnable source code, input parameters,
    software/version information and honest completeness limits.
12. Do not claim a PDF was compiled unless compilation actually succeeded.

PHASE CONTROL:
At the end, provide (i) deliverables, (ii) theorem/proof status, (iii) unresolved
issues, (iv) files created, (v) risks, (vi) a GO/REVISE/STOP recommendation.
Stop and wait for the researcher/ChatGPT audit. Do not begin the next phase
unless explicitly instructed.
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8B. Master audit prompt for ChatGPT

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I am providing the project blueprint and Arena AI's complete output for
the current phase. Act as an independent mathematical referee and research
auditor, not as a supportive proofreader.

TASK:
1. Read the blueprint and all supplied Arena outputs/files.
2. Check every definition, assumption, theorem, proof step, equation,
   parameter formula and computational claim.
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3. Identify invalid reasoning, hidden assumptions, notation conflicts,
   left/right module errors, noncommutative multiplication issues,
   invalid inner products, incorrect duality claims and unsupported
   Gray-map/quantum-code implications.
4. Check the closest literature and distinguish what is already known
   from what may be new. Never call a result novel without evidence.
5. Re-derive important claims independently where possible. If a proof
   is incomplete, say exactly what is missing; do not repair it silently.
6. For code, inspect algorithms and test cases. Distinguish exhaustive
   verification from sampling or heuristic search.
7. Return a structured audit:
   A. Overall verdict: ACCEPT / ACCEPT WITH MINOR REVISIONS /
      MAJOR REVISION / REJECT / INSUFFICIENT EVIDENCE.
   B. Correct results.
   C. Critical mathematical errors.
   D. Missing lemmas/proofs.
   E. Literature and novelty concerns.
   F. Required corrections, prioritized.
   G. Questions requiring the researcher's decision.
   H. GO / REVISE / STOP phase-gate recommendation.
   I. The exact next Arena prompt, but only if progression is justified.

Do not advance to the next phase if foundational assumptions or proofs fail.
Explain the audit in accessible Hindi/Hinglish, but write replacement
mathematics and paper-ready text in formal academic English/LaTeX.
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8C. Prompt for the researcher to request the next Arena phase

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Using the approved audit of the previous phase and the master Research
Blueprint, execute ONLY Phase [INSERT PHASE NUMBER AND TITLE].
First list the accepted assumptions and corrections from the audit.
Do not reopen settled choices unless you find a concrete mathematical
contradiction. Produce every deliverable and proof required by that phase.
At the end, list unresolved issues and stop for independent ChatGPT review.
Do not start the following phase.
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8D. First Arena prompt — Phase 0 only

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Read the complete Research Blueprint. Execute PHASE 0 ONLY:
literature review, novelty assessment and mathematical feasibility.

Search and verify primary sources on:
(i) cyclic codes over noncommutative matrix rings;
(ii) Galois duals and Galois hulls over finite rings and relevant code families;
(iii) quantum codes constructed from matrix-ring codes; and
(iv) quantum codes constructed using Galois hulls.

Deliver:
1. Verified annotated bibliography with full metadata and DOI/URLs.
2. Literature comparison table: ring, code family, duality/hull, construction,
   main theorem/result, limitations and relation to this project.
3. Identify the closest existing papers and state exactly what they already prove.
4. Compare plausible candidate rings M_m(F_q[u]/<u^t>) rather than assuming
   one is best.
5. Assess whether a nontrivial Galois automorphism and a valid non-degenerate
   pairing can exist for each candidate.
6. List mathematical obstacles, especially noncommutativity, left/right
   modules, polynomial centrality, duality and quantum inclusion conditions.
7. Give 2–3 sharply stated candidate research gaps, each with evidence and
   a falsifiable novelty test.
8. Recommend GO / MODIFY / STOP with reasons.
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Do NOT prove later-phase theorems, construct quantum codes, or write the paper.
Do NOT claim novelty based on a title search alone. Stop after Phase 0.
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9. Required project folder and file discipline

Arena should maintain a clear project folder. File names should be stable and versioned; never overwrite the only copy of an approved result.

- 00_BLUEPRINT/Research_Blueprint.docx — this controlling document.
- 01_LITERATURE/ — verified bibliography, paper notes and comparison matrix.
- 02_PHASE_REPORTS/Phase_0/ ... Phase_8/ — one folder per phase.
- 03_THEOREMS/ — theorem register, proof dependencies and proof status.
- 04_COMPUTATION/ — source code, input data, logs and reproducibility notes.
- 05_MANUSCRIPT/ — LaTeX source, bibliography, figures and compiled PDF.
- 06_AUDIT/ — ChatGPT audit reports, corrections and researcher decisions.
- PROJECT_DECISION_LOG.md — date, phase, decision, rationale and approver.
- CHANGELOG.md — every material change in scope, notation, assumptions or results.

10. Mandatory theorem and claim register

For every proposed result, Arena must create a row with: ID; exact statement; assumptions; proof location; dependencies; status (proved / conditional / conjectural / computational only / false); independent audit status; and manuscript section.

ID	Claim / theorem	Assumptions	Proof / file	Dependencies	Status	Audit
To be filled	To be filled	To be filled	To be filled	To be filled	To be filled	To be filled
To be filled	To be filled	To be filled	To be filled	To be filled	To be filled	To be filled
To be filled	To be filled	To be filled	To be filled	To be filled	To be filled	To be filled

11. Quality-control checklist before manuscript submission

- ☐Every definition is unambiguous and consistent with the left/right module convention.
- ☐The chosen matrix ring and coefficient ring are fully specified.
- ☐The Galois automorphism is well-defined and nontrivial where claimed.
- ☐The inner product is well-defined and has the claimed algebraic properties.
- ☐Every duality and hull theorem has a complete proof under stated assumptions.
- ☐Every cyclic-code representation is valid for the chosen noncommutative setting.
- ☐Gray-map properties are proved rather than assumed.
- ☐Quantum construction hypotheses, field-linearity and orthogonality/containment are verified.
- ☐Quantum dimension and distance formulas use consistent notation and conventions.
- ☐Minimum-distance computations are exact or their limitations are explicitly stated.
- ☐All references are real, relevant and bibliographically verified.
- ☐Novelty is supported by comparison with the closest existing results.
- ☐All tables and examples can be reproduced from supplied data/code.
- ☐The final PDF is genuinely compiled and checked if a PDF is supplied.

12. Important mathematical safeguards

- A matrix ring M_m(A_t) is generally noncommutative for m greater than 1. Do not import commutative-ring cyclic-code results without proving they apply.
- An entrywise field automorphism on matrices must be checked against multiplication; an arbitrary matrix operation is not automatically a ring automorphism.
- Do not confuse an additive subgroup, a left module, a right module, a two-sided ideal and a field-linear code.
- Do not assume a standard dot product is suitable for matrix-valued coordinates. Specify a scalar-valued form or an appropriate module-valued orthogonality theory.
- A Galois hull is an intersection C ∩ C-perp_s; its dimension and usefulness depend on the exact ambient module and pairing.
- A Gray map may preserve some properties but not all. State and prove the exact result.
- CSS construction requires appropriate classical linear codes and orthogonality/containment. A hull formula alone is insufficient.
- Check the Quantum Singleton bound $2d \leq n - K + 2$ only after the quantum parameters and distance are correctly established. The defect is $\delta = n + 2 - K - 2d$.
- Do not present computational examples as general theorems. State the tested parameter range and whether enumeration was exhaustive.

13. Final intended outcome

The desired outcome is not simply a long manuscript. It is a defensible mathematical paper containing (i) a genuinely new and correctly defined algebraic framework, (ii) complete structural theorems on Galois duals/hulls, (iii) a valid route to quantum code construction, (iv) reproducible examples and parameter improvements, and (v) a literature-supported novelty claim.

The project may need to change its title, ring, code family or quantum construction after Phase 0/1. Such a change is a sign of rigorous research, not failure. The quality of the final paper takes priority over preserving the initial title.

14. Quick operating instruction

ONE PHASE AT A TIME:
Blueprint → Arena executes one phase → Researcher sends outputs to ChatGPT → ChatGPT audits and writes the next Arena prompt → Researcher approves → Arena continues.

Arena develops. ChatGPT challenges and verifies. Researcher decides.